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Source: *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 2019, Vol. 68, No. 4 (2019), pp. 887-913

Published by: Oxford University Press for the Royal Statistical Society

Stable URL: <https://www.jstor.org/stable/10.2307/26800885>

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Appl. Statist. (2019)
68, Part 4, pp. 887–913

Bayesian inference for stochastic differential equation mixed effects models of a tumour xenography study

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[Received December 2016. Final revision February 2019]

Summary. We consider Bayesian inference for stochastic differential equation mixed effects models (SDEMEMs) exemplifying tumour response to treatment and regrowth in mice. We produce an extensive study on how an SDEMEM can be fitted by using both exact inference based on pseudo-marginal Markov chain Monte Carlo sampling and approximate inference via Bayesian synthetic likelihood (BSL). We investigate a two-compartment SDEMEM, corresponding to the fractions of tumour cells killed by and survived on a treatment. Case-study data consider a tumour xenography study with two treatment groups and one control, each containing 5–8 mice. Results from the case-study and from simulations indicate that the SDEMEM can reproduce the observed growth patterns and that BSL is a robust tool for inference in SDEMEMs. Finally, we compare the fit of the SDEMEM with a similar ordinary differential equation model. Because of small sample sizes, strong prior information is needed to identify all model parameters in the SDEMEM and it cannot be determined which of the two models is the better in terms of predicting tumour growth curves. In a simulation study we find that with a sample of 17 mice per group BSL can identify all model parameters and distinguish treatment groups.

Keywords: Intractable likelihood; Pseudo-marginal Markov chain Monte Carlo sampling; Repeated measurements; State space model; Synthetic likelihood

1. Introduction

Preclinical cancer trials aim at understanding the dynamics of tumour growth and evaluate the effect of treatments such as radiotherapy and chemotherapy in delaying this. A typical trial involves repeated measurements of the volume of solid tumours grown in mice. Tumours are grown until a critical size is reached, in which case the mouse must be sacrificed for ethical reasons, or until a planned end of the study. Data from these trials pose a statistical challenge due to the missing data caused by the sacrifice and the substantial variation in growth patterns between subjects. Even within the same treatment group, some tumours are eliminated following treatment, others continue to grow unaffected and yet others display a decrease in volume followed by regrowth (Laajala *et al.*, 2012).

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0035–9254/19/68887

Heitjan *et al.* (1993) reviewed traditional approaches to analysing tumour xenography experiments, including analysis of variance, multivariate analysis of variance and linear mixed models for tumour volumes. An overall drawback of linear models is that inference targets the mean log-volume of the tumour, which is problematic as all large volumes are censored because of sacrifice. It has been suggested to avoid bias by instead comparing the time to sacrifice, tumour doubling time or a similar survival outcome; see Stuschke *et al.* (1990), Wu and Houghton (2009) and Wu (2011). However, this limits statistical efficiency as the full information in the data is not used. Moreover, delays and doubling times may in practice be difficult to measure accurately because of day-to-day perturbations in growth, measurement error and discrete time follow-up. Rank-based comparisons of composite tumour volume or time-to-sacrifice outcomes are a more robust and powerful approach (Péron *et al.*, 2016). However, this does not offer much insight on the tumour growth dynamics and the effect of the treatment. Also, a fully specified parametric model would be required for power calculations and optimal design.

Demidenko (2013) reviewed non-linear mixed models for tumour growth and regrowth following treatment. In this paper we shall focus on the double-exponential model which identifies two latent compartments corresponding to the fraction of the tumour which is killed by the treatment and the fraction that survives. This model offers a phenomenological explanation for the variation in individual tumour growth patterns, by recognizing

- (a) the proportion of the tumour killed by the treatment,
- (b) the rate of elimination of the dead tumour cells and
- (c) the growth rate of the surviving part of the tumour.

Clinically relevant quantities such as tumour doubling times, tumour growth delay and surviving fraction of tumour cells can be deduced from the double-exponential model (Demidenko, 2006, 2010). More recent non-linear mixed model approaches specify the individual growth curves semiparametrically (Xia *et al.*, 2013), or as splines (Kong and Yan, 2011; Zhao *et al.*, 2011). These models enable much flexibility in individual growth curves but do not share the biological interpretation of the double-exponential or delayed double-exponential models.

A drawback of the double-exponential models, and other classical non-linear mixed models, is that the only source of intrasubject variation is given by independent identically distributed measurement errors. This may not be realistic, as growth rates are subject to day-to-day variation, due to biological processes that are not easily accounted for, e.g. mutations in the cancer and the response of the immune system. In recent years, a number of works have promoted the use of stochastic differential equation mixed effects models (SDEMEmS) as a more realistic alternative to the classical non-linear mixed models. For instance, Donnet *et al.* (2010) found that a stochastic differential equation (SDE) version of the Gompertz growth model is superior to its non-linear deterministic mixed model counterpart, for prediction of the body weight of growing chickens. Donnet and Samson (2013) reported similar findings from pharmacokinetic experiments and Whitaker *et al.* (2017) provide more recent references to inference for SDEMEmS. What prevents more widespread application of SDEMEmS is that inference for non-linear SDE models is overall challenging, even when not considering random effects and measurement errors, because SDEs generally have intractable likelihood functions (Fuchs, 2013). In case measurement error occurs, the observed process is no longer Markovian and a state-space-type model must be considered (Cappé *et al.*, 2006). This paper considers inference for state space models with latent dynamics given by an SDE, which is performed by using state of the art likelihood-based methods employing sequential Monte Carlo (SMC) filters. More specifically, one of our attempts to estimate model parameters applies a pseudo-marginal method (PMM) (Andrieu and Roberts, 2009) which returns exact Bayesian inference, despite

employing SMC approximations. We further consider a methodology which can target more general models beyond the state space class, namely a Bayesian version of the synthetic likelihood (SL) approach (Price *et al.*, 2018). SL was initially proposed in Wood (2010) and does not impose any assumption on the complexity of the model, the only requirement being the ability to simulate artificial data sets from the model. However, SL returns approximate inference, unlike the PMM. To the best of our knowledge this is the first application of SLs to SDEMEMs. We show that Bayesian SL (BSL) applied to SDEMEMs produces results that are qualitatively similar to the exact Bayesian methodology, both with experimental data and with data produced in a simulation study. Further, we find that BSL is easier to calibrate than the PMM and returns results that are less sensitive to the calibration set-up (see the on-line supplementary material). Also, BSL can demonstrate a difference in treatment efficacy in a simulation study with 17 subjects in each treatment group, whereas PMM cannot. Importantly, when considering applications to experimental data, we find that having only 5–8 subjects per treatment group is not sufficient to identify all model parameters accurately. Nevertheless, Bayesian inference still offers an opportunity to perform exploratory data analyses in small-scale experiments, since informative priors based on subject matter expertise may compensate for otherwise too small sample sizes.

Although SDEMEMs are complex models with three layers of randomness, we conclude that BSL should be considered an additional tool for inference in this class of models. Moreover, in future studies BSL would allow us to investigate models beyond the state space type, where SMC methods might not be applicable.

The structure of the paper is as follows: Section 2 contrasts the classical non-linear mixed models with SDEMEMs. Section 3 explains the PMM for exact Bayesian inference in SDEMEMs. Section 4 considers inference by using SLs. In Section 5 we analyse data from a tumour xenography study. In Section 6 results from exact Bayesian inference for ordinary differential equations mixed effects models (ODEMEMs) are given. In Section 7 we run two simulation studies on artificial data. Data and software to run our experiments are available from

<https://rss.onlinelibrary.wiley.com/hub/journal/14679876/series-c-datasets>

(and at the first author's GitHub repository). Further results and methodological considerations are available as on-line supplementary material.

2. Mixed effects models of tumour growth

2.1. Ordinary mixed effects models

Denote with M the number of subjects in a given treatment group. Assume that tumour volumes from subject i are measured at discrete time points $t_{i0} < \dots < t_{in_i}$, $i = 1, \dots, M$. In a planned experiment, such as the experiment that is considered in Section 5, time points will usually be the same for all subjects, i.e. $t_{ij} = t_j$, whereas the number of observations n_i may differ between subjects as the mice are sacrificed when their tumour volume exceeded a critical size prescribed by the ethical guidelines. Denote with $V_{ij} = V_i(t_j)$ the exact tumour volume for subject i at time t_j , $j = 1, \dots, n_i$. We model the observations as

$$Y_{ij} = \log(V_{ij}) + \varepsilon_{ij}, \quad i = 1, \dots, M, \quad j = 1, \dots, n_i, \quad (1)$$

where the ε_{ij} s are independent and identically distributed (IID) normally distributed measurement errors with $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. This means that we assume that tumour volumes are measured with multiplicative log-normal measurement errors. In experimental practice the length and

width of the tumour are measured on the skin surface and the volume is approximated by that of an ellipsoid, resulting in a measurement accuracy that is typically within $\pm 20\%$ of the true volume.

In regulated experiments the mice are sacrificed long before the tumour volumes reach steady state. Hence, unperturbed growth in the control group is adequately described by a simple exponential growth model. Let $\beta_1, \dots, \beta_M$ denote the random subject-specific growth rates; then the growth curves for the control group are given by

$$\frac{dV_i(t)}{dt} = \beta_i V_i(t), \quad V_i(0) = v_{i,0}, \quad i = 1, \dots, M. \quad (2)$$

Distributional assumptions for the β_i s are in Section 2.2. Of course, equation (2) is solved explicitly by $V_i(t) = v_{i,0} \exp(\beta_i t)$. With the further assumption that growth rates are normally distributed and initial tumour volumes are log-normally distributed across the population, the observation model (1) is merely a standard linear mixed model with a random intercept and a random slope.

If tumour volumes are observed post treatment, then the double-exponential model in Demidenko (2013) describes the total volume in terms of surviving tumour cells V_i^{surv} and cells killed by the treatment V_i^{kill} as

$$\left. \begin{aligned} V_i(t) &= V_i^{\text{surv}}(t) + V_i^{\text{kill}}(t), \\ \frac{dV_i^{\text{surv}}(t)}{dt} &= \beta_i V_i^{\text{surv}}(t), \quad V_i^{\text{surv}}(0) = (1 - \alpha_i)v_{i,0}, \quad i = 1, \dots, M, \\ \frac{dV_i^{\text{kill}}(t)}{dt} &= -\delta_i V_i^{\text{kill}}(t), \quad V_i^{\text{kill}}(0) = \alpha_i v_{i,0}. \end{aligned} \right\} \quad (3)$$

Here $\alpha_i \in [0, 1]$ denotes the proportion of the tumour that has been killed by the treatment in subject i , whereas δ_i denotes the elimination rate for the dead tumour cells in subject i . Equation (3) has the explicit solution $V_i(t) = (1 - \alpha_i)v_{i,0} \exp(\beta_i t) + \alpha_i v_{i,0} \exp(-\delta_i t)$. Distributional assumptions for δ_i and α_i are in Section 2.2.

2.2. Stochastic differential equation mixed effects model

The assumption of time constant growth and elimination rates in the ordinary mixed models is usually not realistic, since growth is affected by various biological processes that are not easily accounted for (see Donnet *et al.* (2010) for a motivating example). We therefore suggest replacing the ordinary differential equation model specified by expression (2) with an SDE model such as geometric Brownian motion:

$$dV_i(t) = (\beta_i + \gamma^2/2)V_i(t)dt + \gamma V_i(t)dB_i(t), \quad V_{i,0} = v_{i,0}. \quad (4)$$

Here the $\{B_{i,t}\}_{t \geq 0}$ s are independent standard Brownian motions and γ^2 denotes the intrasubject growth rate variance. This means that the instantaneous growth rate is not exactly β_i but deviates from this by a random normal perturbation. The motivation for including the term $\gamma^2/2$ in the drift of the SDE is that the individual growth process is then given by $V_i(t) = v_{i,0} \exp\{\beta_i t + \gamma B_i(t)\}$ which is a log-normally distributed stochastic process with geometric mean $v_{i,0} \exp(\beta_i t)$, which coincides with the ordinary exponential growth model (2). With the further assumption that growth rates are distributed as $\beta_i \sim \mathcal{N}(\bar{\beta}_0, \sigma_\beta^2)$ and initial tumour volumes as $\log(v_{i,0}) \sim \mathcal{N}(\bar{v}_0, \sigma_0^2)$ across the population, volumes at time t_{ij} would follow a log-normal distribution with geometric mean $\bar{v}_0 \exp(\bar{\beta}_0 t_{ij})$ which is the same as in the ordinary log-linear mixed model. However, in our case-studies we assume that v_{i0} are fixed known mathematical constants, as detailed in Section 5.1.

The ordinary double-exponential model (3) can similarly be replaced by an SDEMEM with specification

$$\left. \begin{aligned} Y_{ij} &= \log(V_{ij}) + \varepsilon_{ij}, & i &= 1, \dots, M, \quad j = 1, \dots, n_i, \\ V_i(t) &= V_i^{\text{surv}}(t) + V_i^{\text{kill}}(t), \\ dV_i^{\text{surv}}(t) &= (\beta_i + \gamma^2/2)V_i^{\text{surv}}(t)dt + \gamma V_i^{\text{surv}}(t)dB_i(t), & V_i^{\text{surv}}(0) &= (1 - \alpha_i)v_{i,0}, \\ dV_i^{\text{kill}}(t) &= (-\delta_i + \tau^2/2)V_i^{\text{kill}}(t)dt + \tau V_i^{\text{kill}}(t)dW_i(t), & V_i^{\text{kill}}(0) &= \alpha_i v_{i,0}, \end{aligned} \right\} \quad (5)$$

with random effects $\beta_i \sim \mathcal{N}(\bar{\beta}, \sigma_\beta^2)$, $\delta_i \sim \mathcal{N}(\bar{\delta}, \sigma_\delta^2)$ and $\alpha_i \sim \mathcal{N}_{[0,1]}(\bar{\alpha}, \sigma_\alpha^2)$ where here and in what follows $\mathcal{N}_{[0,1]}$ denotes a Gaussian distribution truncated to the interval $[0,1]$ (for example, this means that $\alpha_i \sim \mathcal{N}(\bar{\alpha}, \sigma_\alpha^2 | 0 \leq \alpha_i \leq 1)$). The $\{W_i(t)\}_{t \geq 0}$ s are additional standard Brownian motions assumed mutually independent and independent of the $\{B_i(t)\}_{t \geq 0}$ s, of the ε_{ij} , of the (fixed) system initial conditions and of the random effects. Here τ^2 denotes the intrasubject elimination rate variance. The SDEs in model (5) have explicit solutions given by $V_i^{\text{surv}}(t) = V_i^{\text{surv}}(0) \exp\{\beta_i t + \gamma B_i(t)\}$ and $V_i^{\text{kill}}(t) = V_i^{\text{kill}}(0) \exp\{-\delta_i t + \tau W_i(t)\}$ respectively. It is easy to simulate paths from model (5) due to the independent normal increments of the Brownian motions. Note that, whereas processes $\log\{V_i^{\text{surv}}(t)\}$ and $\log\{V_i^{\text{kill}}(t)\}$ both have Gaussian transition densities, instead $\log\{V_i(t)\}$ is not Gaussian distributed, and this prevents an analytic expression for the likelihood function to be found (the integrals in equation (7) in Section 3 cannot be solved analytically). We choose a truncated Gaussian distribution for the individual treatment effect α_i , as this assigns strictly positive probability densities to the values 0 and 1. This is to anticipate that an effective treatment could have the effect that tumours are completely eliminated, whereas an inefficient treatment might not kill any tumour cells.

We stress that the models that were considered above by no means are the only possibilities for specifying dynamical models for tumour growth. Additional random effects could be added to the subject-specific growth curves; for example, to describe a delay in the treatment effect, a different diffusion term could replace $\gamma V_i(t)dB_i(t)$, or a so-called stochastic growth rate model could be specified by $V_i(t) = \exp\{\int_0^t \beta_i(u)du\}$, where the time varying growth rate $\{\beta_i(t)\}_{t \geq 0}$ could be modelled by for example an Ornstein–Uhlenbeck process (a continuous time autoregressive process evolving around its mean $\bar{\beta}_i$). Since the integrated diffusion process in the stochastic growth rate model is not a Markov process, the more general methodology from Section 4 would be needed to analyse this type of model.

3. Likelihood-based inference for stochastic differential equation mixed effects models

In this section we discuss likelihood inference for SDEMEMs such as models (4) and (5) and generalizations thereof. Both models can be viewed as instances of the general state space SDEMEM

$$\left. \begin{aligned} \mathbf{Y}_{ij} &= \mathbf{g}\{\mathbf{X}_i(t_{ij}), \varepsilon_{ij}\}, & \varepsilon_{ij} &\stackrel{\text{IID}}{\sim} \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{I}_{d_y}), \\ d\mathbf{X}_i(t) &= \boldsymbol{\mu}\{\mathbf{X}_i(t), t, \phi_i\}dt + \boldsymbol{\sigma}\{\mathbf{X}_i(t), t, \boldsymbol{\kappa}\}d\mathbf{B}_i(t), & \mathbf{X}_i(t_0) &\sim \pi_0\{\mathbf{x}_i(t_0)|\phi_i\}, \\ \phi_i &\stackrel{\text{IID}}{\sim} p(\phi_i|\boldsymbol{\eta}), \end{aligned} \right\} \quad (6)$$

where each $\mathbf{Y}_{ij}$ has dimension $\dim(\mathbf{Y}_{ij}) = d_y$, $\dim(\varepsilon_{ij}) = d_y$, $\mathbf{I}_{d_y}$ is the $d_y \times d_y$ identity matrix and $\mathbf{X}_i(t)$ has dimension $\dim\{\mathbf{X}_i(t)\} = d_x$, with $d_x \geq d_y$ at every t . Model (6) has the following interpretation: for each subject i , $\{\mathbf{X}_i(t)\}_{t \geq 0}$ represents the hidden (unobservable) biological process of interest, with dynamics governed by the drift and diffusion functions $\boldsymbol{\mu}(\cdot)$

and $\sigma(\cdot)$ which are assumed known, save from the subject-specific parameters (random effects) ϕ_i and the common model parameters κ . In the case of model (5), the latent process is $\mathbf{X}_i(t) = (V_i^{\text{kill}}, V_i^{\text{surv}})$, $\kappa = (\gamma, \tau)$, $\phi_i = (\log(\alpha_i), \log(\beta_i), \log(\delta_i))$, $\eta = (\bar{\alpha}, \bar{\beta}, \bar{\delta}, \sigma_\alpha, \sigma_\beta, \sigma_\delta)$ and $\mathbf{X}_i(t_0) = (x_{i,0}^{\text{surv}}, x_{i,0}^{\text{kill}})$ with $x_{i,0}^{\text{surv}} = (1 - \alpha_i)v_{i,0}$ and $x_{i,0}^{\text{kill}} = \alpha_i v_{i,0}$. Regularity conditions for the existence and uniqueness of a solution to the SDE can be found for example in Fuchs (2013). Observations $\{\mathbf{Y}_{ij}\}$ are assumed to consist of discrete time measurements of the latent process $\{\mathbf{X}_i(t)\}$, perturbed with measurement error ε_{ij} via a known function $\mathbf{g}(\cdot)$. For example model (5) is specified with $\mathbf{g}(\mathbf{v}, \varepsilon) = \log(v^{\text{kill}} + v^{\text{surv}}) + \varepsilon$. Finally, the subject-specific random effects ϕ_i are assumed distributed with a density $p(\cdot|\eta)$ parameterized by the ‘population parameter’ η . The aim of our analysis is to perform inference for the vector parameter $\theta = (\eta, \kappa, \sigma_\varepsilon)$. A main feature of the SDEM (6) is its ability to discriminate between the temporal intrasubject variability κ , the intersubject variability $\text{var}(\phi_i)$ and the measurement error variance σ_ε^2 . Knowledge of these distinct sources of variation will be valuable when planning experiments and performing power calculations.

It is important to note that measurements $\mathbf{Y}_{ij}$ are conditionally independent given the latent states $\mathbf{X}_{ij} := \mathbf{X}_i(t_j)$ and ϕ_i , implying that model (6) is a state space model (Cappé *et al.*, 2006). This, as well as the Markov property of the latent process $\{\mathbf{X}_i(t)\}$, is essential for the inference methods that are described in this section to work. However, these are not required properties for the methodology in Section 4.

Denote with $\mathbf{y}_i = \{\mathbf{y}_{ij}\}_{j=1, \dots, n_i}$ the collection of observations for subject i and with $\mathbf{X}_i = \{\mathbf{X}_{ij}\}_{j=1, \dots, n_i}$ the corresponding values of the latent process. Let $\mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_M) \in \mathcal{Y}$ denote the full set of measurements for all subjects in a certain experimental group. Standard methods for frequentist as well as Bayesian estimation of the model parameters θ require the evaluation of the likelihood function $p(\mathbf{y}|\theta) = \prod_{i=1}^M p(\mathbf{y}_i|\theta)$. The hidden Markov structure implies the following derivation:

$$\begin{aligned} p(\mathbf{y}_i|\theta) &= \int p(\mathbf{y}_i|\phi_i; \theta) p(\phi_i|\theta) d\phi_i \\ &= \int \left\{ \int p(\mathbf{y}_i|\mathbf{X}_i; \theta) p(\mathbf{X}_i|\phi_i; \theta) d\mathbf{X}_i \right\} p(\phi_i|\theta) d\phi_i \\ &= \int \left[\int \left\{ \prod_{j=1}^{n_i} p(\mathbf{y}_{ij}|\mathbf{X}_{ij}, \theta) p(\mathbf{X}_{i,j}|\mathbf{X}_{i,j-1}, \phi_i; \theta) \right\} p(\mathbf{X}_{i0}|\phi_i, \theta) d\mathbf{X}_i \right] p(\phi_i|\theta) d\phi_i. \end{aligned} \quad (7)$$

The term $p(\mathbf{X}_{i0}|\phi_i, \theta)$ vanishes in either of the cases where $\mathbf{X}_{i0}$ is included among the random effects or is assumed to be a known constant: $\mathbf{X}_{i0} := \mathbf{x}_{i0}$.

Function (7) is not analytically tractable. Thus, inference for SDEM (6) relies on either more specific model assumptions, such as the latent processes being Gaussian, or the use of computationally intensive methods. Delattre and Lavielle (2013) showed how to conduct likelihood inference in SDEM (6) by using the stochastic approximate expectation–maximization algorithm (SAEM) coupled with an extended Kalman filter. Donnet and Samson (2014) proposed a particle Markov chain Monte Carlo algorithm to perform the ‘stochastic’ step in SAEM. In either case the use of SAEM requires explicit specification of sufficient summary statistics for the augmented likelihood $p(\mathbf{y}_i, \mathbf{X}_i, \phi_i|\theta)$. Although providing fast and accurate inference in models with a latent Gaussian structure, the derivation of the summary statistics is a tedious if not impossible task for more complex models of realistic interest. See Picchini (2018) for a likelihood-free version of SAEM.

Bayesian inference targets the parameter posterior distribution $\pi(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta)\pi(\theta)$ where $\pi(\theta)$ is the corresponding prior distribution. Bayesian methodology for SDEM (6) was first

Table 1. Algorithm 1: a pseudo-marginal MCMC algorithm

Step 1: input, a positive integer R ; fix a starting value θ^* or generate it from its prior $\pi(\theta)$ and set $\theta_1 := \theta^*$; set a kernel $q(\theta'|\theta)$; use algorithm 2 or the auxiliary particle filter to obtain an unbiased estimate $\hat{p}(\mathbf{y}|\theta^*)$ of $p(\mathbf{y}|\theta^*)$; set $r = 1$
Output, R correlated draws from $\pi(\theta|\mathbf{y})$ (possibly after a burn-in)
Step 2: generate a $\theta^\# \sim q(\theta^\#|\theta^*)$; use algorithm 2 or the auxiliary particle filter to obtain an unbiased estimate $\hat{p}(\mathbf{y}|\theta^\#)$ of $p(\mathbf{y}|\theta^\#)$
Step 3: generate a uniform random draw $u \sim U(0, 1)$, and calculate the acceptance probability

$$\alpha = \min \left[1, \frac{\hat{p}(\mathbf{y}|\theta^\#) q(\theta^*|\theta^\#) \pi(\theta^\#)}{\hat{p}(\mathbf{y}|\theta^*) q(\theta^\#|\theta^*) \pi(\theta^*)} \right]$$

If $u > \alpha$, set $\theta_{r+1} := \theta_r$; otherwise set $\theta_{r+1} := \theta^\#$, $\theta^* := \theta^\#$ and $\hat{p}(\mathbf{y}|\theta^*) := \hat{p}(\mathbf{y}|\theta^\#)$; set $r := r + 1$ and go to step 4
Step 4: if $r \leq R$ repeat steps 2–3; otherwise stop

Table 2. Algorithm 2: SMC BF for mixed effects state space models

Input, a positive integer L , a starting value for θ and a starting value $\mathbf{x}_0$; set time $t_0 = 0$ and corresponding starting states $\mathbf{X}_{i0} = \mathbf{x}_0$ (we use the convention that all steps involving the index l must be performed for all $l \in \{1, \dots, L\}$)
Output, all the $\hat{p}(\mathbf{y}_{ij}|\mathbf{y}_{i1:j-1})$, $i = 1, \dots, M$, $j = 1, \dots, n_i$
for $i = 1, \dots, M$ do
draw $\phi_i^l \sim p(\phi_i|\theta)$
if $j = 1$ then
sample $\mathbf{x}_{i1}^l \sim p(\mathbf{x}_{i1}|\mathbf{x}_{i0}, \phi_i^l; \theta)$
compute $w_{i1}^l = p(\mathbf{y}_{i1}|\mathbf{x}_{i1}^l)$ and $\hat{p}(\mathbf{y}_{i1}) = \sum_{l=1}^L w_{i1}^l / L$
normalization, $\tilde{w}_{i1}^l := w_{i1}^l / \sum_{l=1}^L w_{i1}^l$; interpret $\tilde{w}_{i1}^l$ as a probability associated with $\mathbf{x}_{i1}^l$
resampling, sample L times with replacement from the probability distribution $\{\mathbf{x}_{i1}^l, \tilde{w}_{i1}^l\}$; denote the sampled particles with $\tilde{\mathbf{x}}_{i1}^l$
end if
for $j = 2, \dots, n_i$ do
forward propagation, sample $\mathbf{x}_{ij}^l \sim p(\mathbf{x}_{ij}|\tilde{\mathbf{x}}_{i,j-1}^l, \phi_i^l; \theta)$
compute $w_{ij}^l = p(\mathbf{y}_{ij}|\mathbf{x}_{ij}^l)$ and normalize $\tilde{w}_{ij}^l := w_{ij}^l / \sum_{l=1}^L w_{ij}^l$
compute $\hat{p}(\mathbf{y}_{ij}|\mathbf{y}_{i1:j-1}) = \sum_{l=1}^L \tilde{w}_{ij}^l / L$
resample L times with replacement from $\{\mathbf{x}_{ij}^l, \tilde{w}_{ij}^l\}$; sampled particles are $\tilde{\mathbf{x}}_{ij}^l$
end for
end for

studied by Donnet *et al.* (2010) who implemented a Gibbs sampler that applies to the case where the SDE has an explicit solution, and which can be extended to the more general state space model by using an Euler–Maruyama discretization. A recent review of Bayesian inference methods for SDEMEds can be found in Whitaker *et al.* (2017). It is important to note that Markov chain Monte Carlo algorithms can be constructed to sample from the exact posterior of θ , for models admitting a non-negative unbiased estimator of $p(\mathbf{y}|\theta)$ (Beaumont, 2003; Andrieu and Roberts, 2009). In this spirit, we exemplify a PMM (Andrieu and Roberts, 2009) by using SMC methods. The key idea is to substitute the intractable $p(\mathbf{y}|\theta)$ with an unbiased non-negative estimate $\hat{p}(\mathbf{y}|\theta)$, and to plug this in an otherwise standard Metropolis–Hastings algorithm (algorithm 1 in Table 1). An unbiased estimate of the likelihood function can be obtained by using SMC filters, of which two popular examples are the bootstrap filter (BF) (Gordon *et al.*, 1993), adapted in algorithm 2 in Table 2, and the auxiliary particle filter (APF) (Pitt and Shephard, 1999; Pitt *et al.*, 2012), with a version that is suitable for our case-studies detailed

in the on-line supplementary material. Here we describe the BF, as it is more approachable for a general audience and sufficient to convey the methodology. The interested reader is referred to the supplementary material which contains useful notes on how to implement either a BF or an APF for model (5), as well as a comparison between the BF and APF. In most cases of practical interest, the forward propagation step in both the BF and the APF requires a numerical scheme, such as Euler–Maruyama discretization (see for example Golightly and Wilkinson (2011)), though this is not so with model (5) as the analytic solutions for $V_i^{\text{surv}}(t)$ and $V_i^{\text{kill}}(t)$ are known (Section 2.2). The approximated likelihood is

$$\hat{p}(\mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^M \hat{p}(\mathbf{y}_i|\boldsymbol{\theta}), \quad (8)$$

and when using BF we have

$$\hat{p}(\mathbf{y}_i|\boldsymbol{\theta}) = \hat{p}(\mathbf{y}_{i1}|\boldsymbol{\theta}) \prod_{j=2}^{n_i} \hat{p}(\mathbf{y}_{ij}|\mathbf{y}_{i,1:j-1}, \boldsymbol{\theta}) = \prod_{j=1}^{n_i} \left(\frac{1}{L} \sum_{l=1}^L w_{ij}^l \right), \quad (9)$$

where L is the number of particles that are used to propagate the latent state forward and the w_{ij}^l s are importance weights. Strategies to tune the value of L can be found in Doucet *et al.* (2015) and Sherlock *et al.* (2015). In the context of our case-study, results by using different values of L are compared in the supplementary material. We performed the resampling step by using the stratified method of Kitagawa (1996). Also, note that in Sections 5–7 we use a Gaussian kernel $q(\cdot|\cdot)$ to propose parameters via the adaptive Gaussian random-walk algorithm of Haario *et al.* (2001).

The main distinction between model (6) and other state space models is that latent states are subject specific and can be further decomposed into a time-dependent component $\mathbf{X}_i$ and a time-independent component ϕ_i . Therefore, when applying SMC filters we first draw ϕ_i and then, conditionally on such a draw, we propagate forward particles corresponding to the states $\mathbf{X}_i$. As proven by Del Moral (2004) and Pitt *et al.* (2012), each individual estimate $\hat{p}(\mathbf{y}_i|\boldsymbol{\theta})$ produced by a BF and APF is unbiased (where the expectation is taken with respect to the distribution that is used to generate all the random variables employed in the SMC approximation). Since measurements from different subjects are assumed independent, it follows that $\hat{p}(\mathbf{y}|\boldsymbol{\theta})$ in likelihood (8) is an unbiased estimator for $p(\mathbf{y}|\boldsymbol{\theta})$. Because of this, parameters drawn according to algorithm 1 have stationary distribution $\pi(\boldsymbol{\theta}|\mathbf{y})$ (after a burn-in), for any number of particles L (Beaumont, 2003; Andrieu and Roberts, 2009).

4. Approximate inference for stochastic differential equation mixed effects model by using synthetic likelihoods

In this section we discuss approximate Bayesian inference for SDEMEmS by using synthetic likelihoods (Wood, 2010). Similarly to approximate Bayesian computation (ABC) (see Marin *et al.* (2012) for a review), the SL methodology is a black box approach solely relying on simulations from the assumed data-generating model. It is therefore a tool that is suitable for models having an otherwise intractable likelihood. It is important to note that neither ABC nor SL requires the model to have a state space representation. Compared with exact methods, the drawbacks of the approximate methodologies is the loss of statistical efficiency and a need to validate their performance case by case.

Similarly to ABC, SL relies on a set of carefully selected summary statistics for the data $\mathbf{s} := \mathbf{s}(\mathbf{y})$. However, whereas in ABC no assumption is made for the distribution of $\mathbf{s}$, SL assumes that

summary statistics follow a multivariate normal distribution: $\mathbf{s} \sim \mathcal{N}\{\boldsymbol{\mu}(\boldsymbol{\theta}), \boldsymbol{\Sigma}(\boldsymbol{\theta})\}$ (see Fasiolo *et al.* (2018) and An *et al.* (2018) on relaxing this assumption). If this holds true, and if parameters in $\boldsymbol{\theta}$ can be identified from $\boldsymbol{\mu}(\boldsymbol{\theta})$ and $\boldsymbol{\Sigma}(\boldsymbol{\theta})$, then inference for $\boldsymbol{\theta}$ can be based on the Gaussian likelihood of $\mathbf{s}$ instead of the intractable likelihood of $\mathbf{y}$. Most often $\boldsymbol{\mu}(\boldsymbol{\theta})$ and $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ are unknown functions which must be approximated with simulations. Hence synthetic likelihoods can be viewed as an instance of the simulated method of moments (McFadden, 1989).

The implementation of SL is straightforward. For a given $\boldsymbol{\theta}$, N synthetic data sets $\mathbf{y}^{*1}, \dots, \mathbf{y}^{*N}$ are generated independently from the model; hence each entry in the vector $\mathbf{y}^{*n}$ belongs to the space $\mathcal{Y}$ (see the notation in Section 3) and $\dim(\mathbf{y}^{*n}) = \dim(\mathbf{y})$, $n = 1, \dots, N$. Summary statistics $\mathbf{s}^{*n} = \mathbf{s}(\mathbf{y}^{*n})$ are computed for each simulated data set and from these we obtain the estimates

$$\hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta}) = \frac{1}{N} \sum_{n=1}^N \mathbf{s}^{*n},$$

$$\hat{\boldsymbol{\Sigma}}_N(\boldsymbol{\theta}) = \frac{1}{N-1} \sum_{n=1}^N (\mathbf{s}^{*n} - \hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta}))(\mathbf{s}^{*n} - \hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta}))'.$$

When applying SL to SDEMEmS it is important that summary statistics reflect the hierarchical structure of the model, i.e. to anticipate the intraindividual and interindividual variation of the experimental data we construct subject-specific summaries $\mathbf{s}_i := \mathbf{s}^{\text{intra}}(\mathbf{y}_i)$ for $i = 1, \dots, M$, as well as summaries that represent interindividual variation between all subjects $\mathbf{s}^{\text{inter}} := \mathbf{s}^{\text{inter}}(\mathbf{y}_1, \dots, \mathbf{y}_M)$, so that $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_M, \mathbf{s}^{\text{inter}})$. The summaries that were used in our case-study are described in Section 5.2.

Approximate normality for summary statistics can often be argued theoretically, by appealing to the central limit theorem. If the sample size is small or the summary statistics do not admit a central limit theorem, then the normal assumption would have to be verified empirically by using simulations; see Wood (2010) for details.

Here we follow Price *et al.* (2018), who proposed a fully Bayesian approach, henceforth referred to as BSL. A BSL procedure samples from the exact posterior $\pi(\boldsymbol{\theta}|\mathbf{s})$ without incurring any bias caused by a finite N (note that ‘exact’ sampling is ensured only if the distribution of $\mathbf{s}$ is really Gaussian). The key feature exploits the idea underlying the PMM discussed in Section 3, where an unbiased estimator is used in place of the unknown likelihood function. Price *et al.* (2018) noted that plugging the estimates $\hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta})$ and $\hat{\boldsymbol{\Sigma}}_N(\boldsymbol{\theta})$ into the Gaussian likelihood $p(\mathbf{s}|\boldsymbol{\theta})$ in general results in a biased estimator $p_N(\mathbf{s}|\boldsymbol{\theta})$ of $p(\mathbf{s}|\boldsymbol{\theta})$. However, this can be avoided by instead adopting the unbiased estimator of Ghurye and Olkin (1969):

$$\hat{p}(\mathbf{s}|\boldsymbol{\theta}) = (2\pi)^{-d/2} \frac{c(d, N-2)}{c(d, N-1)(1-1/N)^{d/2}} |(N-1)\hat{\boldsymbol{\Sigma}}_N(\boldsymbol{\theta})|^{-(n-d-2)/2}$$

$$\times \left[\psi \left\{ (N-1)\hat{\boldsymbol{\Sigma}}_N(\boldsymbol{\theta}) - \frac{(\mathbf{s} - \hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta}))(\mathbf{s} - \hat{\boldsymbol{\mu}}_N(\boldsymbol{\theta}))'}{1-1/N} \right\} \right]^{(N-d-3)/2}. \quad (10)$$

Here π denotes the mathematical constant (not the prior), $d = \dim(\mathbf{s})$, N is assumed to satisfy $N > d + 3$, and for a square matrix $\mathbf{A}$ the function $\psi(\mathbf{A})$ is defined as $\psi(\mathbf{A}) = |\mathbf{A}|$ if $\mathbf{A}$ is positive definite and $\psi(\mathbf{A}) = 0$ otherwise, where $|\mathbf{A}|$ is the determinant of $\mathbf{A}$. Finally

$$c(k, v) = 2^{-kv/2} \pi^{-k(k-1)/4} \prod_{i=1}^k \Gamma\left\{\frac{1}{2}(v-i+1)\right\}.$$

To produce inference for SDEMEmS with BSL we can use algorithm 3 in Table 3, by analogy with algorithm 1 in Section 3. The former merely uses SL to draw from the posterior $\pi(\boldsymbol{\theta}|\mathbf{s})$

Table 3. Algorithm 3: BSL

<i>Input</i> , a positive integer R ; the observed summary statistics $\mathbf{s}$: fix a starting value θ^* or generate it from the prior $\pi(\theta)$; set $\theta_1 = \theta^*$; choose a kernel $q(\theta' \theta)$; set $r = 1$
<i>Output</i> , R correlated samples from $\pi(\theta \mathbf{s})$
<i>Step 1</i> : conditionally on θ^* generate independently N summaries $\mathbf{s}^{*1}, \dots, \mathbf{s}^{*N}$, compute moments $\hat{\mu}_N(\theta^*)$, $\hat{\Sigma}_N(\theta^*)$ and $\hat{p}(\mathbf{s} \theta^*)$ from estimator (10)
<i>Step 2</i> : generate a $\theta^\# \sim q(\theta^\# \theta^*)$; conditionally on $\theta^\#$ generate independently $\mathbf{s}^{\#1}, \dots, \mathbf{s}^{\#N}$, compute $\hat{\mu}_N(\theta^\#)$, $\hat{\Sigma}_N(\theta^\#)$ and $\hat{p}(\mathbf{s} \theta^\#)$
<i>Step 3</i> : generate a uniform random draw $u \sim U(0, 1)$, and calculate the acceptance probability
$\alpha = \min \left[1, \frac{\hat{p}(\mathbf{s} \theta^\#) q(\theta^* \theta^\#) \pi(\theta^\#)}{\hat{p}(\mathbf{s} \theta^*) q(\theta^\# \theta^*) \pi(\theta^*)} \right]$
If $u > \alpha$, set $\theta_{r+1} := \theta_r$; otherwise set $\theta_{r+1} := \theta^\#$, $\theta^* := \theta^\#$ and $\hat{p}(\mathbf{s} \theta^*) := \hat{p}(\mathbf{s} \theta^\#)$; set $r := r + 1$ and go to step 4
<i>Step 4</i> : repeat steps 2–3 as long as $r \leq R$

instead of $\pi(\theta|\mathbf{y})$, assuming that $\mathbf{s}$ follows a Gaussian distribution. In our case-studies, we use algorithm 3 to estimate the model parameters of the SDEMEMs (4) and (5). To the best of our knowledge, this is the first application of the SL methodology to SDEMEMs. We refer to Section 5.2 and the on-line supplementary material for further considerations on the implementation of BSL.

One of the main conclusions that we draw from analysing experimental data (Section 5) is that the sample sizes are too small to obtain accurate inference for all model parameters. Most importantly, it is difficult to identify the mean treatment efficacy $\bar{\alpha}$ between different treatment groups with both BSL and the PMM. Interestingly, our simulation study (Section 7) suggests that, with a moderately larger sample size, BSL can identify $\bar{\alpha}$ whereas the PMM cannot.

5. Case-study

We consider data from a tumour xenography study originally including four treatment groups (1, chemotherapy; 2, radiation therapy; 3, combination therapy I; 4, combination therapy II), and one untreated control (group 5). Each group consists of 7–8 mice. Mice were followed up on Mondays, Wednesdays and Fridays for six consecutive weeks or until their tumour volume exceeded 1000 mm³, in which case the mouse was sacrificed as prescribed by the Danish legislation for the use of animals in scientific research. In groups 2 and 4 about half of the mice were sacrificed within the treatment period or shortly after, and because of the reduced sample sizes these will not be considered any further.

Treatment in equal size doses was applied on days 1, 4 and 6 of the study. Afterwards no treatment was administered. The repeated measurements of tumour volumes in the three remaining groups are shown in Fig. 1. It is obvious that growth patterns vary substantially between subjects. In the untreated control group a single mouse with a slowly growing tumour survived for 32 days before sacrifice whereas all other untreated mice were sacrificed within 10 days. In the active treatment groups we see patterns of decay followed by regrowth which match the characteristic shape of the double-exponential curve. In the same groups we also see tumours that appear to grow continuously, unaffected by the treatment. An outlying mouse in group 1 appears to present a slowly vanishing tumour. Most likely the implanted tumour cells never grew to form a tumour in this mouse in the first place; hence its data were excluded from the analyses. Several mice display tumour volumes that are stable over shorter durations of time. These stable

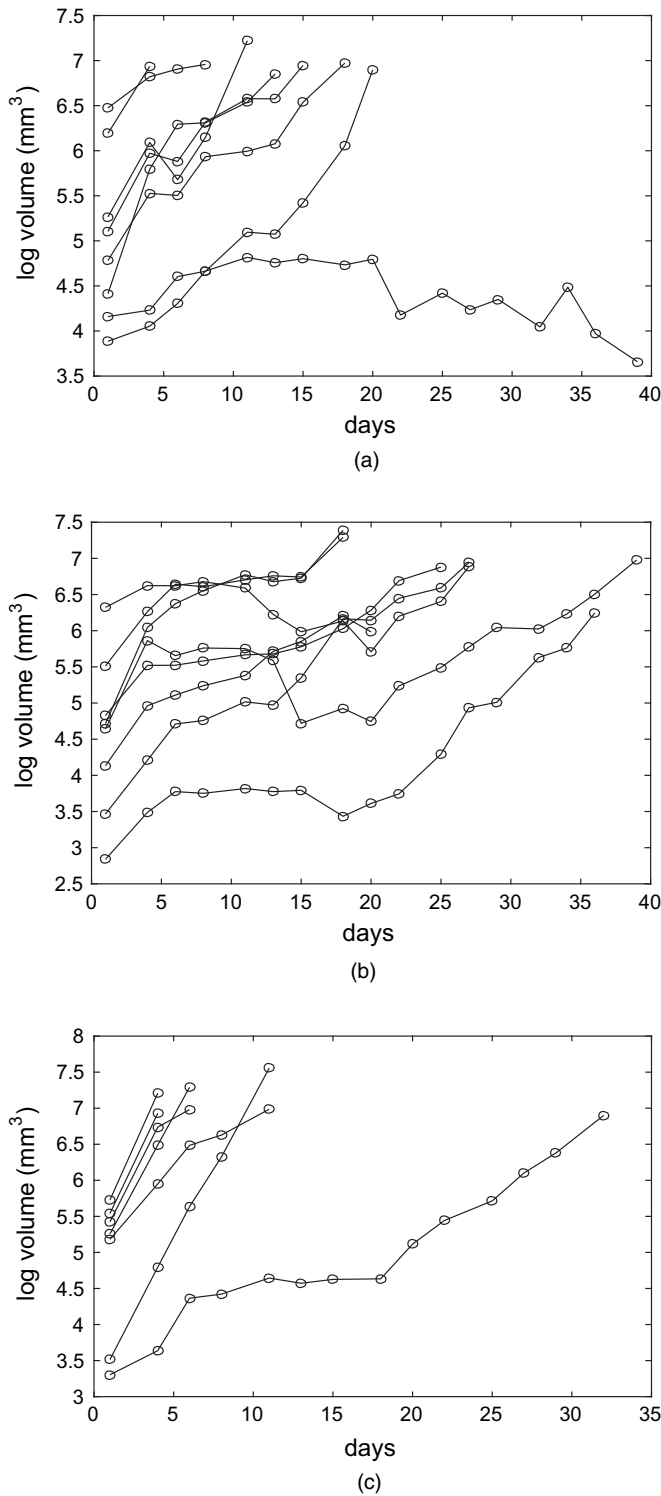


Fig. 1. Log-volumes data for three treatment groups: (a) group 1; (b) group 3; (c) group 5

periods deviate from the growth patterns of the ordinary simple and double-exponential mixed models but can be explained by the random variations in growth and decay rates which are modelled in the double-exponential SDEM. We applied the double-exponential SDEM (5) to model the post-treatment log-volumes, i.e. starting from (and including) day 6. Separate model fits were obtained for treatment groups 1 and 3. Tumour growth in the untreated controls (group 5) was modelled with the simple exponential SDEM (4) starting from day 1 of the study.

For Bayesian analysis of the double-exponential SDEM (5) we choose a truncated Gaussian prior on the average treatment effect $\bar{\alpha} \sim \mathcal{N}_{[0,1]}(0.6, 0.2^2)$. Note that this assigns strictly positive probabilities to the values 0 and 1. This is to anticipate that an effective treatment could have the effect that tumours are completely eliminated, whereas an inefficient treatment might not kill any tumour cells at all. For the remaining parameters in model (5) we choose priors $\log(\beta) \sim \mathcal{N}(0.7, 0.6^2)$, $\log(\delta) \sim \mathcal{N}(0.7, 0.6^2)$, $\sigma_\beta \sim \text{InvGam}(4, 2)$, $\sigma_\delta \sim \text{InvGam}(4, 2)$, $\sigma_\alpha \sim \text{InvGam}(5, 1.5)$, $\gamma \sim \text{InvGam}(5, 7)$, $\tau \sim \text{InvGam}(5, 7)$ and $\sigma_\varepsilon \sim \text{InvGam}(2, 1)$, where $\text{InvGam}(a, b)$ denotes the inverse gamma distribution with shape a and scale b . Note that positive model parameters have been reparameterized by their logarithms. We refer to the on-line supplementary material for BSL results obtained by using less informative priors. To enhance numerical stability the observational times were scaled as $t_{ij} = t_{ij}/t_{\max} \in [0, 1]$, since $t_{\max} = 39$ days was the maximum time of follow-up after which the remaining mice were sacrificed. Parameter estimates should be interpreted accordingly. Software and data are available from <https://rss.onlinelibrary.wiley.com/hub/journal/14679876/series-c-datasets> (and at the first author's GitHub repository).

5.1. Results by using exact Bayesian inference

We fitted model (5) separately for groups 1 and 3 by using the PMM algorithm, as described in Section 3. Unbiased estimates of the likelihood function were obtained via the APF (as found in the on-line supplementary material). For each subject, $v_{i0} = y_{i1}$ was considered a known constant. Recall, however, that in model (5) initial states v_{i0}^{surv} and v_{i0}^{kill} depend also on α_i . Algorithm 1 was initialized at $\log(\bar{\beta}) = 1.6$, $\log(\bar{\delta}) = 1.6$, $\log(\bar{\alpha}) = -0.36$, $\log(\gamma) = 0$, $\log(\tau) = 0$, $\log(\sigma_\beta) = -0.7$, $\log(\sigma_\delta) = -0.7$, $\log(\sigma_\alpha) = -2.3$ and $\log(\sigma_\varepsilon) = 0$. We used $L = 2000$ particles and $L_2 = 5$ (the number of particles propagated from each of the L particles to compute first-stage weights; see the supplementary material). Chains of length $R = 20000$ were produced and the computation took about 53 min for treatment group 1 ($M = 5$), and 167 min for treatment group 3 ($M = 8$) with MATLAB (Mathworks, 2017) code running on a Intel Core i7-4790, 3.60-GHz computer. For both groups, average acceptance rates observed during the execution of algorithm 1 were equal to 30%. The chains' convergence was verified by using the scale reduction factor $\hat{R}$ (Gelman and Rubin, 1992) as implemented in R's coda package (Plummer *et al.*, 2006). We considered three chains initialized at very dispersed values compared with the marginal posteriors. All values for $\hat{R}$ were below 1.1 except for σ_α ($\hat{R} = 1.2$); hence the chains appear to have converged.

Results are shown in Table 4 (the initial 10000 draws were discarded as burn-in). The treatment efficacy is estimated at $\bar{\alpha} = 60\%$ in group 3 and at $\bar{\alpha} = 52\%$ in group 1; however, the corresponding posteriors are very wide in both groups. Thus it is not possible to draw conclusions on differences in treatment efficacy between the two groups (the posterior marginal of the difference of the two efficacies is not shown). In Section 7.1 we show that having larger sample sizes enables a much better identification of the treatment's efficacy $\bar{\alpha}$ if inference is made with BSL. Note that posteriors for $\log(\beta)$, γ , τ and σ_ε are informative when compared with their priors. Also, the

estimate for $\bar{\beta}$ is higher in group 1 than in group 3, as it should be by looking at Fig. 1 (recall that in group 1 the decaying growth curve was excluded before analysis). It is reassuring that the measurement error variance is estimated consistently as $\hat{\sigma}_\varepsilon \simeq 0.1\text{--}0.2$ in all groups. This means that tumour volumes were measured with a relative accuracy approximately within $\pm 20\%$, which is realistic for an experiment of this type. In contrast, we found the marginal posterior for σ_α to be highly sensitive to the choice of its prior, i.e. the posterior distribution followed the shape of the prior, regardless of the choice of hyperparameters.

The one-compartment model (4) was fitted to the untreated controls (group 5). The priors were the same as for the corresponding parameters in the two-compartment model (5). Parameter estimates are shown in Table 4. Estimates of the mean population growth rate $\bar{\beta}$ are higher than for groups 1–3, and the diffusion coefficient γ is also higher than the corresponding stochasticities for the two-compartment models, but the credibility intervals are wide. The measurement error standard deviation σ_ε is compatible with the previous model fits, which is reassuring.

To make a rough assessment of whether model (5) is realistic compared with the data, we simulated growth curves by using the posterior means estimated with the PMM (Table 4). The simulations are shown in Figs 2 and 3. The overall impression is that model (5) is capable of generating growth dynamics that are similar to the experimental data. A more thorough comparison is produced by using posterior predictive checks in Section 5.2.3.

A comparison between the PMM by using BFs *versus* APFs is presented in the on-line supplementary material. We found that results obtained with the APF are more stable. This is no surprise as the BF is known to degenerate when the measurement error is small (in our case the typical log-volumes are more than an order of magnitude larger than σ_ε). Moreover we compared the PMM with BSL (see in addition the next section), finding that results from the PMM seem sensitive with the number of particles L , whereas BSL is less affected by the number of simulated data sets N .

Table 4. Posterior means and 95% posterior intervals†

Parameter	Results for group 1	Results for group 3	Results for group 5
$\bar{\beta}$	5.81 [3.82,7.83]	3.33 [2.07,4.64]	6.70 [4.09,8.90]
$\bar{\delta}$	6.59 [4.90,8.75]	3.93 [2.93,5.04]	7.48 [6.20,8.98]
$\bar{\alpha}$	1.84 [0.68,4.59]	1.14 [0.40,2.32]	—
γ	1.90 [0.53,4.69]	1.52 [0.43,3.68]	—
τ	0.52 [0.24,0.84]	0.60 [0.31,0.91]	—
σ_β	0.41 [0.13,0.74]	0.47 [0.17,0.84]	—
σ_δ	1.13 [0.66,1.71]	1.09 [0.70,1.52]	1.49 [1.07,2.09]
σ_α	1.03 [0.63,1.42]	0.92 [0.56,1.32]	1.64 [1.26,2.15]
σ_ε	1.50 [0.68,2.98]	1.82 [1.02,2.63]	—
σ_β	1.51 [0.71,2.77]	1.75 [1.03,2.64]	—
σ_δ	0.61 [0.23,1.37]	0.51 [0.19,1.67]	0.68 [0.23,1.96]
σ_α	0.55 [0.25,1.19]	0.59 [0.23,1.28]	0.40 [0.27,0.60]
σ_ε	0.67 [0.24,1.68]	0.76 [0.26,2.23]	—
σ_β	0.66 [0.23,1.74]	0.71 [0.25,1.91]	—
σ_δ	0.37 [0.16,0.74]	0.29 [0.15,0.48]	—
σ_α	0.30 [0.14,0.68]	0.43 [0.14,1.16]	—
σ_ε	0.22 [0.19,0.30]	0.20 [0.19,0.23]	0.23 [0.20,0.31]
σ_ε	0.17 [0.10,0.28]	0.11 [0.07,0.17]	0.18 [0.11,0.29]

†For each parameter we first report exact Bayesian inference by using the PMM and then approximate inference by using BSL.

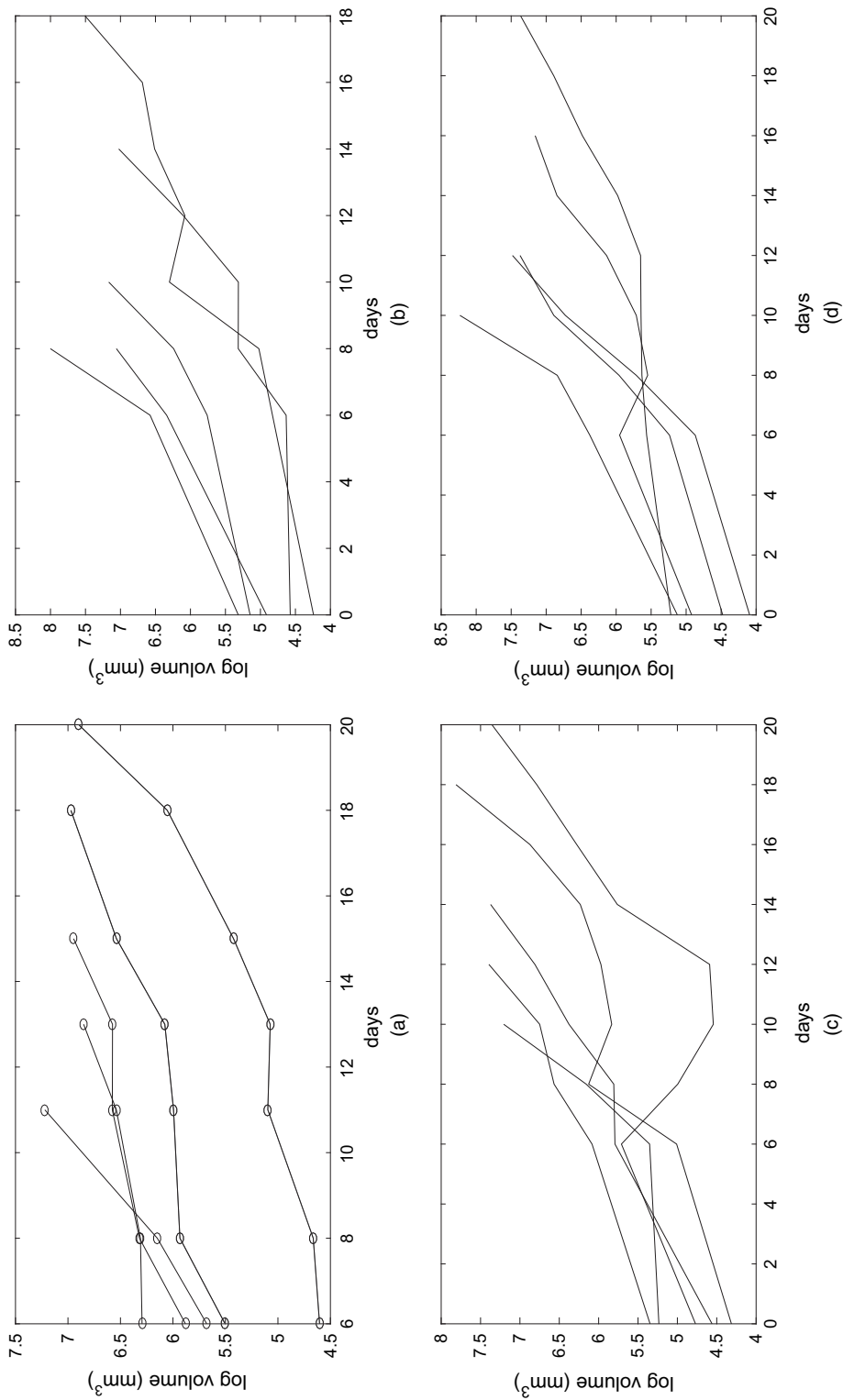


Fig. 2. (a) Fitted data in group 1 and (b), (c), (d) three realizations from model (5) estimated with exact Bayesian methodology: (a) does not report data for one excluded mouse; recall that for this group measurements at days 1 and 4 were disregarded during estimation; hence times on the abscissa start at day 6

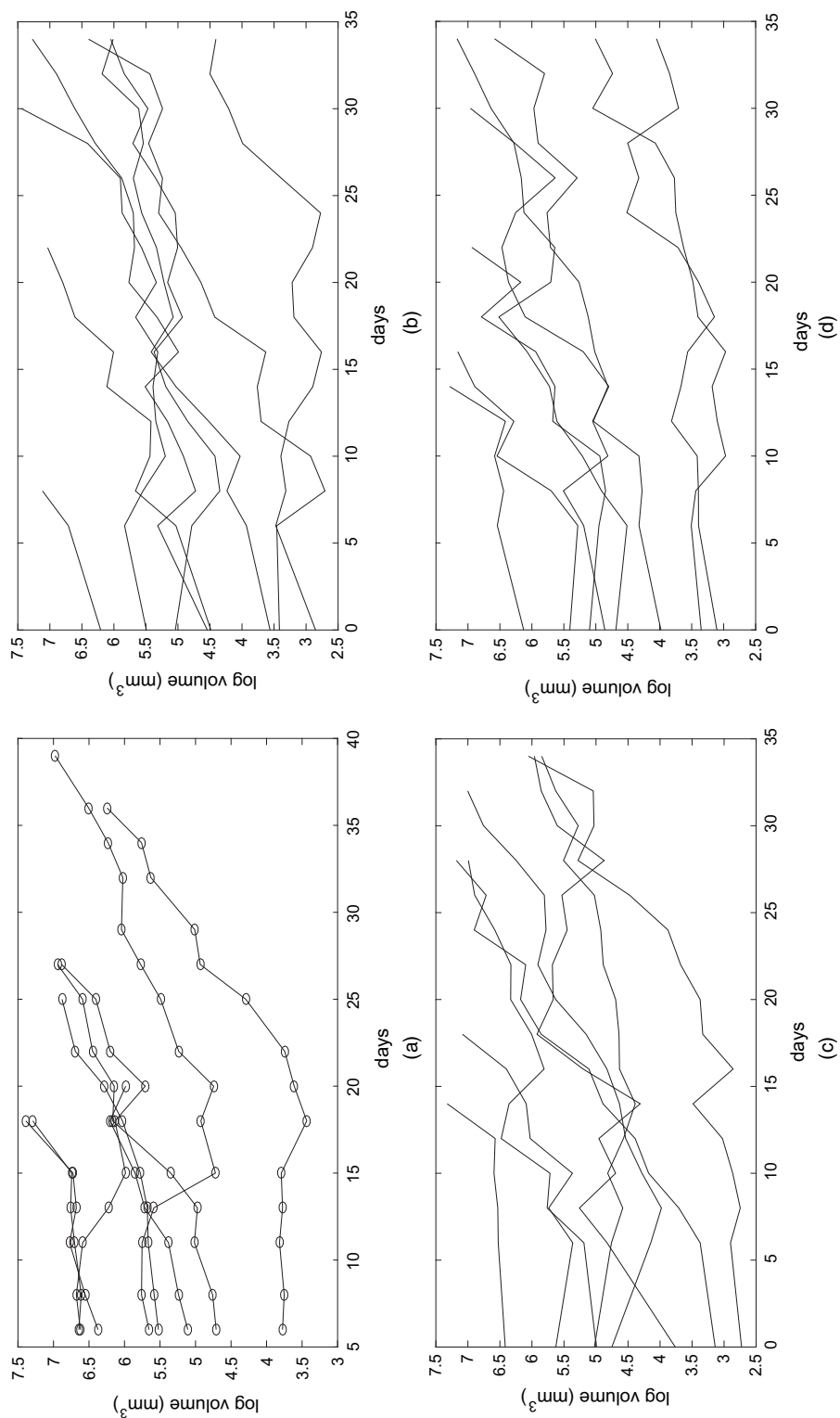


Fig. 3. (a) Fitted data in group 3 and (b), (c), (d) three realizations from model (5) estimated with exact Bayesian methodology: recall that for this group measurements at days 1 and 4 were disregarded during estimation; hence times on abscissa start at day 6

5.2. Results by using synthetic likelihoods

The application of the SL methodology from Section 4 relies on a selection of summary statistics. Therefore, before presenting any results, we first describe how our chosen summary statistics were constructed.

5.2.1. Summary statistics

We first define the components for each individual (vector) summary $\mathbf{s}_i := \mathbf{s}^{\text{intra}}(\mathbf{y}_i)$:

- (a) the mean absolute deviation for the repeated measurements $\text{MAD}\{y_{ij}\}_{j=1:n_i}$;
- (b) the slope of the line segment connecting the first and the last observation, $\{y_i(t_{n_i}) - y_i(t_1)\}/(t_{n_i} - t_1)$;
- (c) the value of the first measurement y_{i1} ;
- (d) the value of the second measurement y_{i2} ;
- (e) the estimated slope of a first-order auto-regressive fit of the repeated measurements, i.e. $\hat{\beta}_{i1}$ from the regression $E(y_{ij}) = \beta_{i0} + \beta_{i1}y_{i,j-1}$.

Note that, when fitting model (4) to the control group, the last summary statistic was dropped to prevent $\Sigma_N(\theta)$ from becoming singular (several mice had only two observations so the second and fifth summary were perfectly correlated). Additional interindividual (population) summary statistics $\mathbf{s}^{\text{inter}}$ are

- (a) $\text{MAD}\{y_{i1}\}_{i=1:M}$, the mean absolute deviation between subjects at the first time point (day 6 for the active treatment groups and day 1 for the control group);
- (b) the same as in (a) but for the second time point;
- (c) the same as in (a) but for the last time point.

Therefore when fitting group 3 ($M = 8$ subjects) the total vector of summaries $\mathbf{s}$ contains 43 features, since we have five features per subject plus three interindividual features. In the absence of previous literature considering the construction of summary statistics for SDEMEmS, our custom-made summaries follow commonsense intuition. For example it seems reasonable to include in $\mathbf{s}^{\text{intra}}(\mathbf{y}_i)$ a robust measure of variability, MAD, for individual trajectories. Also, since the overall behaviour of the trajectories is increasing, we believe that the slope of the line connecting first and last observations can give insight on the volume growth rate. Similarly, the values of the first two individual measurements could represent an assessment of the initial growth. The first-order auto-correlation is a standard measure of information in dynamic models. Similarly, we assess variation between subjects by using $\mathbf{s}^{\text{inter}}$: for this we assess the variation between trajectories at several sampling times, in our case by using MAD on measurements from three different times points.

5.2.2. Results by using Bayesian synthetic likelihood

We use $N = 3000$ simulated data sets to construct the SL approximation at each value of θ and run $R = 20000$ iterations of the BSL algorithm 3, as described in Section 4. For group 5, because of the small number of subjects, we doubled N to $N = 6000$ (otherwise we obtain a very variable SL, whose occasional overestimation causes stickiness in the chains). For comparability, we adopted the same priors and initial parameter values as in the exact Bayesian analysis. During the execution of the algorithm we observed an acceptance rate of about 30% and the procedure required about 520 s for group 3 ($M = 8$ subjects). We checked the convergence of three chains initialized at the same dispersed values used to assess the convergence of the PMM. For each parameter the $\hat{R}$ -value was below 1.04; hence the chains are converging.

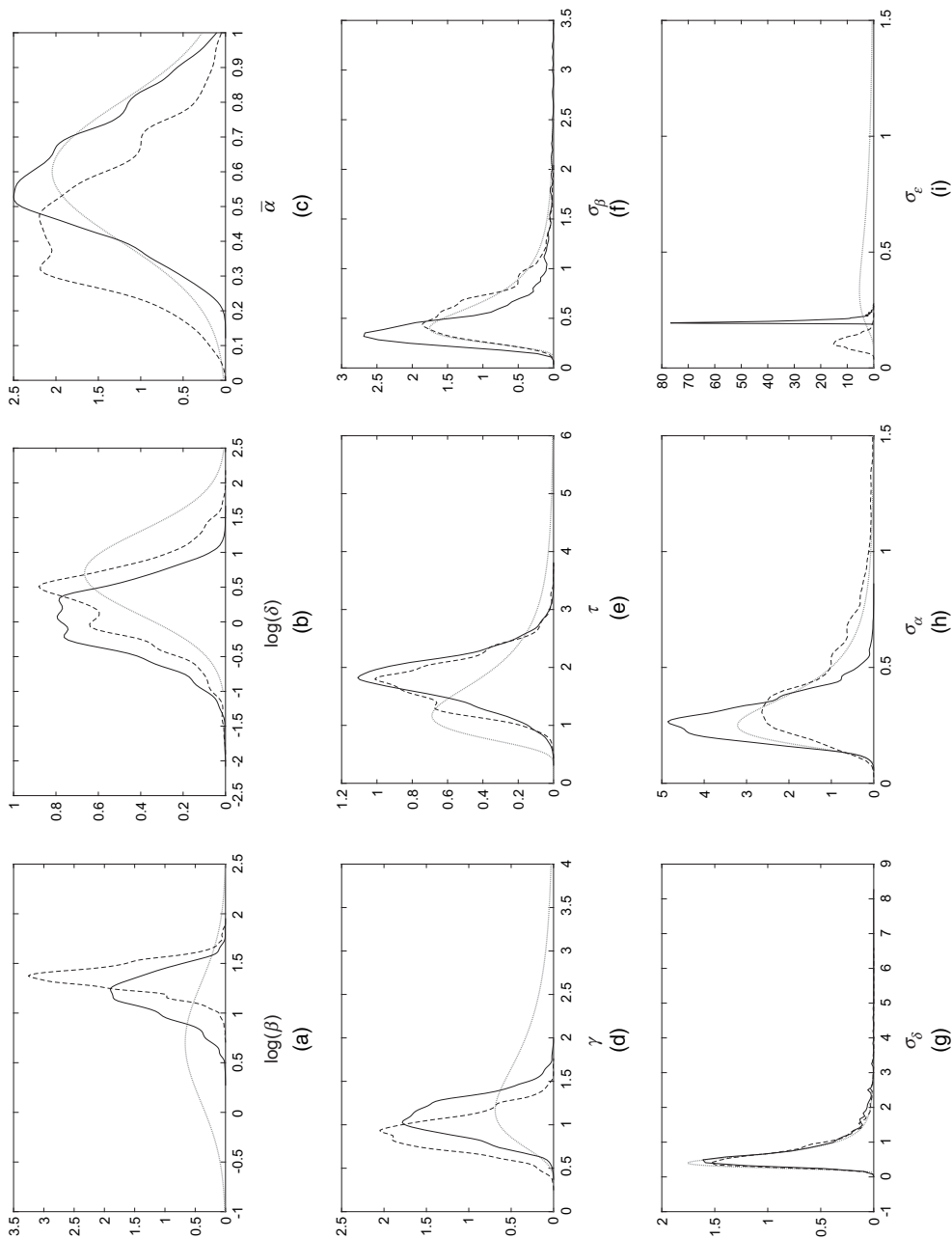


Fig. 4. Treatment group 3, exact posteriors via the PMM by using the APF (—), SL posteriors (---) and prior densities (.....) (the prior density for σ_ϵ was multiplied by 4 for ease of display): (a) $\log(\beta)$; (b) $\log(\delta)$; (c) $\bar{\alpha}$; (d) γ ; (e) τ ; (f) σ_β ; (g) σ_δ ; (h) σ_α ; (i) σ_ϵ

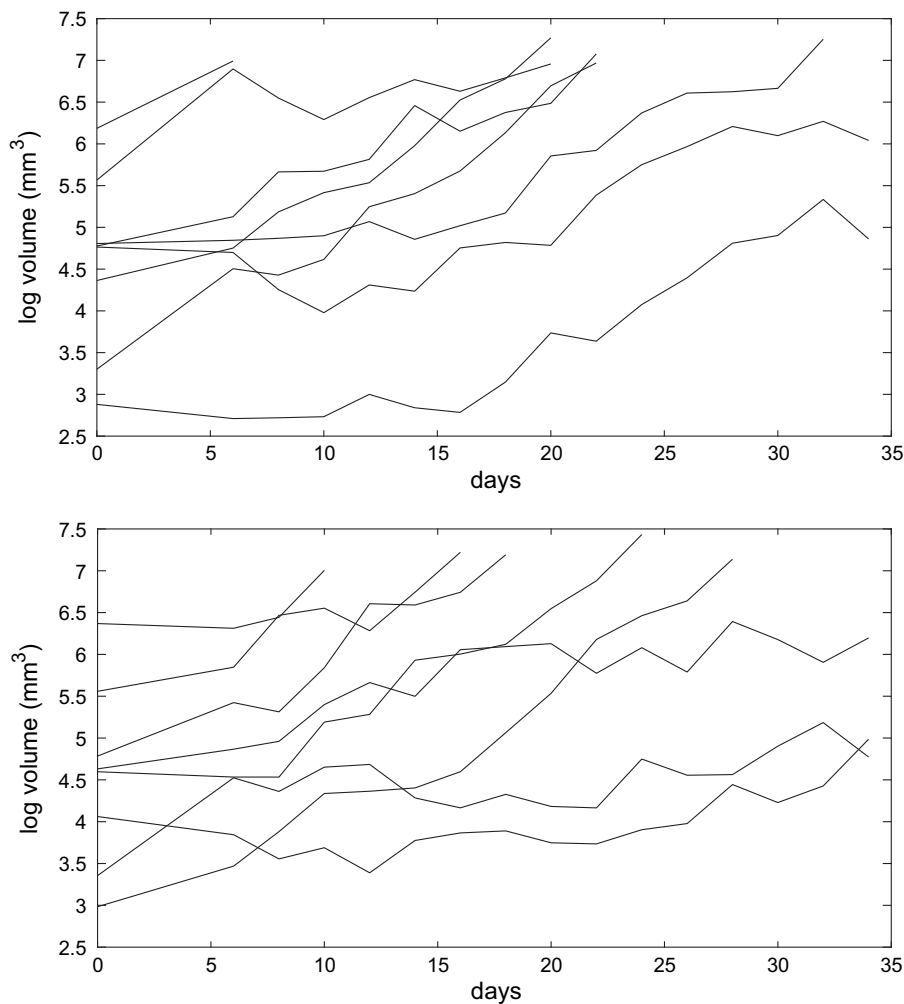


Fig. 5. Group 3: two realizations from model (5) estimated with SL

Posterior estimates obtained with BSL (discarding a burn-in of 10000 iterations) are compared with the PMM in Table 4, and the approximate marginals for group 3 are compared in Fig. 4. Although some differences in posterior means are found between BSL and the PMM, these appear to be of minor consequence to the anticipated growth patterns. Fig. 5 shows simulated growth curves (based on the posterior means estimated with BSL) that are overall similar to Fig. 3, which was obtained by using the corresponding PMM estimates. A more careful evaluation of the BSL model fit is carried out by using posterior predictive checks in Section 5.2.3 below. All the selected summary statistics were found to be approximately normally distributed (see the on-line supplementary material).

Additional analyses reported in the supplementary material show that BSL returns results that are closer to those from the PMM, provided that the PMM is run with a larger number of particles (say, $L = 3000$) and moreover that BSL is less sensitive to the specific choice of N than the PMM is to the choice of L . Furthermore we investigated the effect of using less informative priors for $\log(\delta)$, σ_β and σ_δ . The conclusion is that the volume of data considered

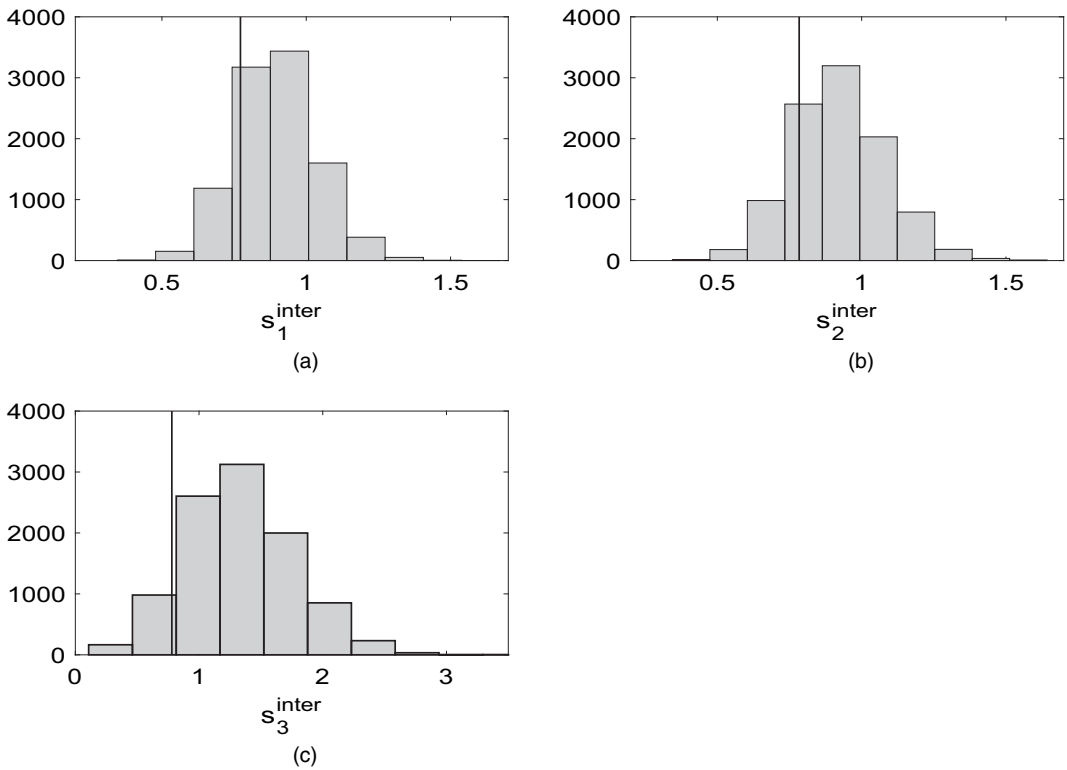


Fig. 6. Posterior predictive checks for group 3 generated from BSL: simulated intersubject summaries (a) s_1^{inter} , (b) s_2^{inter} and (c) s_3^{inter} (|, values for the corresponding statistics from the observed data)

is not sufficiently informative for these parameters, i.e. the information that is carried by the model cannot depart from the prior information for σ_β and σ_δ . The (log-)elimination rate $\log(\bar{\delta})$ does depart from its prior, but at the expense of increased variability.

Comparing computational times between BSL and the PMM in a fair way is difficult, since the two algorithms have a completely different structure. Both methods perform similarly in terms of the acceptance rate (30% in both cases); however, in terms of raw numbers, the PMM is clearly more intensive since at each Markov chain Monte Carlo iteration the model is simulated $L \times L_2 = 10000$ times, whereas for BSL the model is simulated only $N = 3000$ times (plus the overhead time that is needed to compute summary statistics out of each simulated trajectory). Given this, 1000 Markov chain Monte Carlo iterations using data from group 3 required 8.35 min with the PMM and 0.44 min with BSL.

5.2.3. Posterior predictive checks

Posterior predictive checks were made for data in group 3, following the reasoning and notation in the on-line supplementary material. We used the 10000 draws that were produced as output of the BSL algorithm 3 (after burn-in) to simulate corresponding 10000 independent sets of summaries from the posterior predictive distribution. Next, these were compared with the observed summaries from the experimental data. Since $M = 8$ subjects are considered, each $\mathbf{s}^*$ contains 43 summaries. Fig. 6 shows the histograms for the three intersubject summaries whereas the intra-subject summaries for subject 1 are shown in Fig. 7. Corresponding plots for the remaining sub-

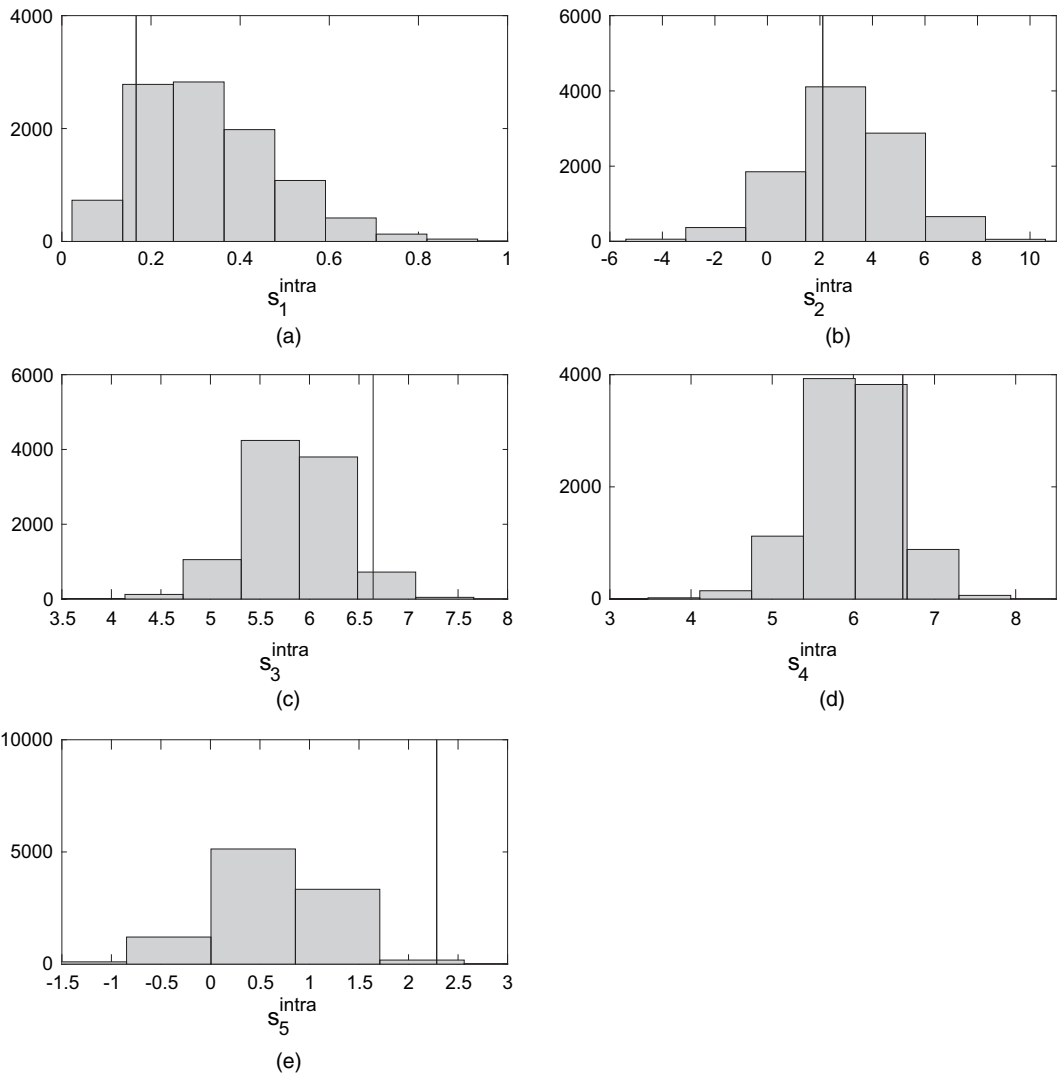


Fig. 7. Posterior predictive checks for group 3 generated from BSL: simulated intrasubject summaries for (subject 1) (a) s_1^{intra} , (b) s_2^{intra} , (c) s_3^{intra} , (d) s_4^{intra} and (e) s_5^{intra} (|, values for the corresponding statistics from the observed data)

jects can be found in the on-line supplementary material. Regarding the intersubject summaries, the model generates summaries that are comparable with the corresponding observed summaries. Regarding the intrasubject summaries for subject 1, the estimated model complies well with the observed summaries, except for s_5^{intra} which seems less plausible. However, for the other subjects, the observed s_5^{intra} is highly probable under $p(s^*|s)$ (see the supplementary material).

6. Comparison with an ordinary differential equation mixed effects model

Although SDEMEmS may have a better chance to capture real life within- and between-subjects variability, inference is complicated by the need to make delicate implementation and calibration

Table 5. Posterior means and 95% posterior intervals from exact Bayesian inference for the ODEMEmS

<i>Parameter</i>	<i>Results for group 1</i>	<i>Results for group 3</i>
$\bar{\beta}$	9.93 [6.90,13.30]	5.71 [3.43,7.48]
$\bar{\delta}$	2.24 [0.68,5.58]	1.77 [0.63,4.02]
$\bar{\alpha}$	0.42 [0.08,0.80]	0.46 [0.09,0.82]
σ_{β}	0.69 [0.23,2.01]	1.87 [0.47,3.81]
σ_{δ}	0.59 [0.22,1.44]	0.54 [0.22,1.22]
σ_{α}	0.36 [0.14,0.91]	0.46 [0.16,1.15]
σ_{ε}	0.76 [0.56,1.04]	0.60 [0.51,0.72]

decisions for the simulations set-up, to approximate their likelihoods. In contrast, ODEMEmS have tractable likelihood functions and can be analysed with off-the-shelf software utilizing robust MCMC inference via Hamiltonian Monte Carlo methods (Betancourt, 2017), where exploration of the posterior surface is guided by exact gradients.

For a comparison of the two model types we have fitted the ODEMEm (1)–(3) separately to data from treatment groups 1 and 3. Parameters of interest are $\theta = (\bar{\beta}, \bar{\delta}, \bar{\alpha}, \sigma_{\beta}, \sigma_{\delta}, \sigma_{\alpha})$. Inference based on 20000 draws from the true posterior $\pi(\theta|\mathbf{y})$ is presented in Table 5 (see the on-line supplementary material for details). We note that the estimated mean growth and decay rates ($\bar{\beta}$ and $\bar{\delta}$ respectively) from the ODEMEmS are higher compared with those estimated from the SDEMEmS. Importantly, estimated residual variation σ_{ε} here is 3–4 times larger than for the SDEMEmS. Also, marginal posteriors are wider for all the parameters. In particular, identification of $\bar{\alpha}$ remains elusive in both groups. By analogy with Section 5.2.3 we performed posterior predictive checks to evaluate the fit of the ODEMEm for group 3. Results are shown in the supplementary material. These indicate a better fit for subject 1 than that obtained by using SDEMEmS.

However, larger sample sizes (or more frequent measurements over time) are needed to determine which model performs best in terms of both in-sample and out-of-sample predictions.

7. Simulation studies

We have conducted a small-scale simulation study to investigate the statistical properties of the PMM and BSL in the context of the SDEMEm (5).

30 data sets were generated independently from the model with parameters θ_0 set to the posterior means that were obtained with exact Bayesian methodology on group 3, as found in Table 4 except for $\bar{\alpha}$ which was set to 0.75 for consistency with the simulation study in Section 7.1. Each simulated data set has measurements for $M = 8$ subjects, with observational times identical to those of group 3, and growth curves initiated at the same values $v_{i,0}$ as in the previous sections. To each of the generated data sets we applied both the PMM and BSL, initializing the algorithms at the same starting parameters as in previous analyses and using the same set-up as in the case-study, where the PMM uses $L = 2000$ particles and $L_2 = 5$ and the BSL approach uses $N = 3000$. We collected the 30 posterior means $\hat{\theta}_b$ and computed their median biases and root-mean-square errors $\text{RMSE} = \sqrt{\{\sum_{b=1}^{30} (\hat{\theta}_b - \theta_0)'(\hat{\theta}_b - \theta_0)/30\}}$. Results obtained with the PMM and BSL are similar, except for $\bar{\delta}$; see Table 6 and Figs 8 and 9. Fig. 9 considers boxplots of the difference of the posterior means biases, so a positive difference for a

Table 6. Simulation study with $M = 8$ subjects: true parameter values θ_0 , median bias and RMSE by using the PMM and BSL

Method	Statistic	$\bar{\beta}$	$\bar{\delta}$	$\bar{\alpha}$	γ	τ	σ_{β}	σ_{δ}	σ_{α}	σ_{ε}
θ_0 PMM	Bias	3.33	1.14	0.75	1.09	1.82	0.51	0.76	0.29	0.20
	RMSE	1.09	0.60	0.21	0.27	0.51	0.039	0.213	0.045	0.129
BSL	Bias	-0.93	0.29	-0.28	-0.26	-0.67	-0.039	-0.213	0.001	0.143
	RMSE	1.07	0.44	0.30	0.27	0.62	0.053	0.216	0.033	0.161

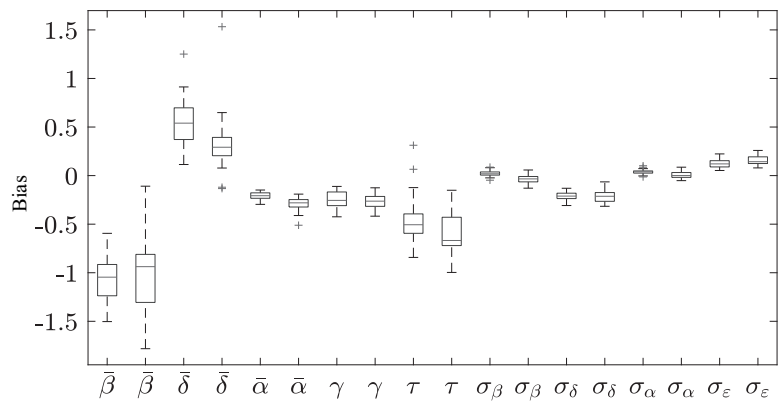


Fig. 8. Simulation study with $M = 8$ subjects: boxplots of the bias of 30 posterior means obtained with the PMM and BSL; starting from the left-hand side, bias of $\bar{\beta}$ obtained via the PMM, then the bias of $\bar{\beta}$ obtained via BSL, and so on

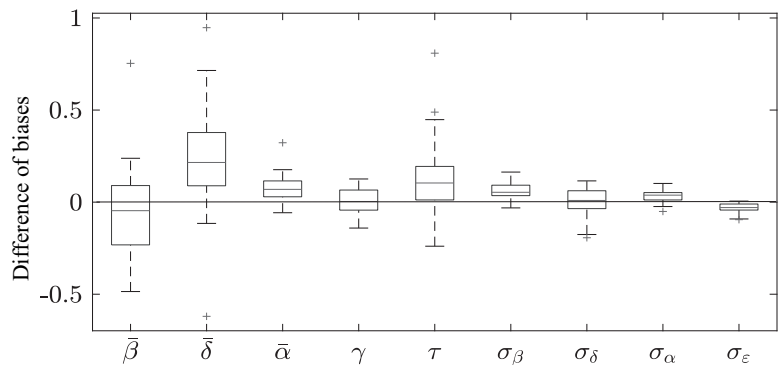


Fig. 9. Simulation study with $M = 8$ subjects: boxplots of the difference of the biases shown in Fig. 8, namely $\text{bias(PMM)} - \text{bias(BSL)}$ (——, difference of biases equal to 0)

given parameter implies that the bias is larger (in absolute value) when the PMM is used. Fig. 9 seems to suggest a slightly better performance of BSL compared with the PMM. However, the number of repetitions $B = 30$ is too small to be conclusive. Unfortunately, performing a larger simulation study would be computationally very intensive. Running only 30 simulations required about 41 h with the PMM and 7 h with BSL. In the next section we explore the effect of increasing the number of subjects for a single experiment.

7.1. Results by using larger sample sizes

To investigate whether the problems that we had in identifying the model parameters cease if the sample size is increased, we simulated two data sets corresponding to two groups, each containing $M = 17$ subjects but having different treatment efficacies. The model parameters for the first group, $\mathcal{D}_1$, was set to the same values as for the PMM estimates for group 1 in Table 4, except for $\bar{\alpha}$, here set to $\bar{\alpha} = 0.35$ (low treatment efficacy). The model parameters for the second group, $\mathcal{D}_2$, was set to the same values as for the PMM estimates for group 3 in Table 4, except for $\bar{\alpha}$, here set to $\bar{\alpha} = 0.75$ (high treatment efficacy). We applied both BSL and the PMM to analyse the data, initializing the algorithms at the same starting values as in previous sections. Because of the increased sample size (and hence a larger spread of data points) we use a larger number of simulations $N = 6000$ with BSL whereas for the PMM we use $L = 5000$ and $L_2 = 10$. The posteriors are in Fig. 10.

7.1.1. Bayesian synthetic likelihood results

Fig. 10 shows that the mean growth rates $\bar{\beta}$ for the surviving tumour cells are correctly identified, with a higher growth rate for $\mathcal{D}_1$ than for $\mathcal{D}_2$, and the two posteriors for $\bar{\beta}$ are well separated. Compared with the smaller sample size $M = 8$ (Fig. 4) the mean treatment efficacy $\bar{\alpha}$ is much better identified. Note that the posterior for $\bar{\alpha}$ in $\mathcal{D}_2$ shows a better identification of the ground truth parameter than in $\mathcal{D}_1$. This is most likely due to the longer trajectories in $\mathcal{D}_2$ (subjects survive longer). The separation between the two marginal posteriors suggests that we could obtain more accurate inferences for treatments efficacy in real data by using BSL. We further note that the residual variability σ_ε is difficult to identify with high precision. Given that for log-normal data the coefficient of variation is given by $\sqrt{\{\exp(\sigma_\varepsilon^2) - 1\}}$, and, since the four marginals in Fig. 10(a) suggest an estimate $\hat{\sigma}_\varepsilon \approx 0.3$, we obtain an estimated coefficient of variation of about 0.31. The true coefficient of variation equals 0.20.

7.1.2. Pseudo-marginal method results

The most notable difference between BSL and the PMM is that the PMM cannot identify $\bar{\alpha}$ both for $\mathcal{D}_1$ and for $\mathcal{D}_2$. The posteriors for the remaining model parameters are overall similar. Given that BSL is an approximate methodology, some differences between the two methods are expected. We believe that the failure to identify $\bar{\alpha}$ points to the need of improving the way that the PMM algorithm is constructed. Currently the random effects ($\log(\alpha_i)$, $\log(\beta_i)$, $\log(\delta_i)$) are simulated from their *unconditional* distributions, e.g. $\log(\alpha_i) \sim \mathcal{N}_{[0,1]}(\bar{\alpha}, \sigma_\alpha^2)$, instead of being simulated from a distribution *conditional* on data. How to construct such distributions is left for future research, but clearly ‘blind’ simulation of α_i comes at a cost, as this enters the starting conditions for the dynamics in equation (5). BSL also propagates random effects blindly, but it explicitly encodes information on the population variability via the s^{inter} -statistics.

8. Summarizing remarks

We have introduced a new mixed effects model for the analysis of repeated measurements of tumour volumes in mice in a tumour xenography study. For each subject the dynamics for the exact, unobservable, tumour volumes are modelled by SDEs, whereas observed volumes are assumed perturbed with measurement error. The resulting model is an SDEM, which is of state space type. SDEMs provide a very useful representation for repeated measurement data since they can distinguish several sources of variability, in the present example: intraindividual temporal variability, biologic variability between subjects and measurement error (residual

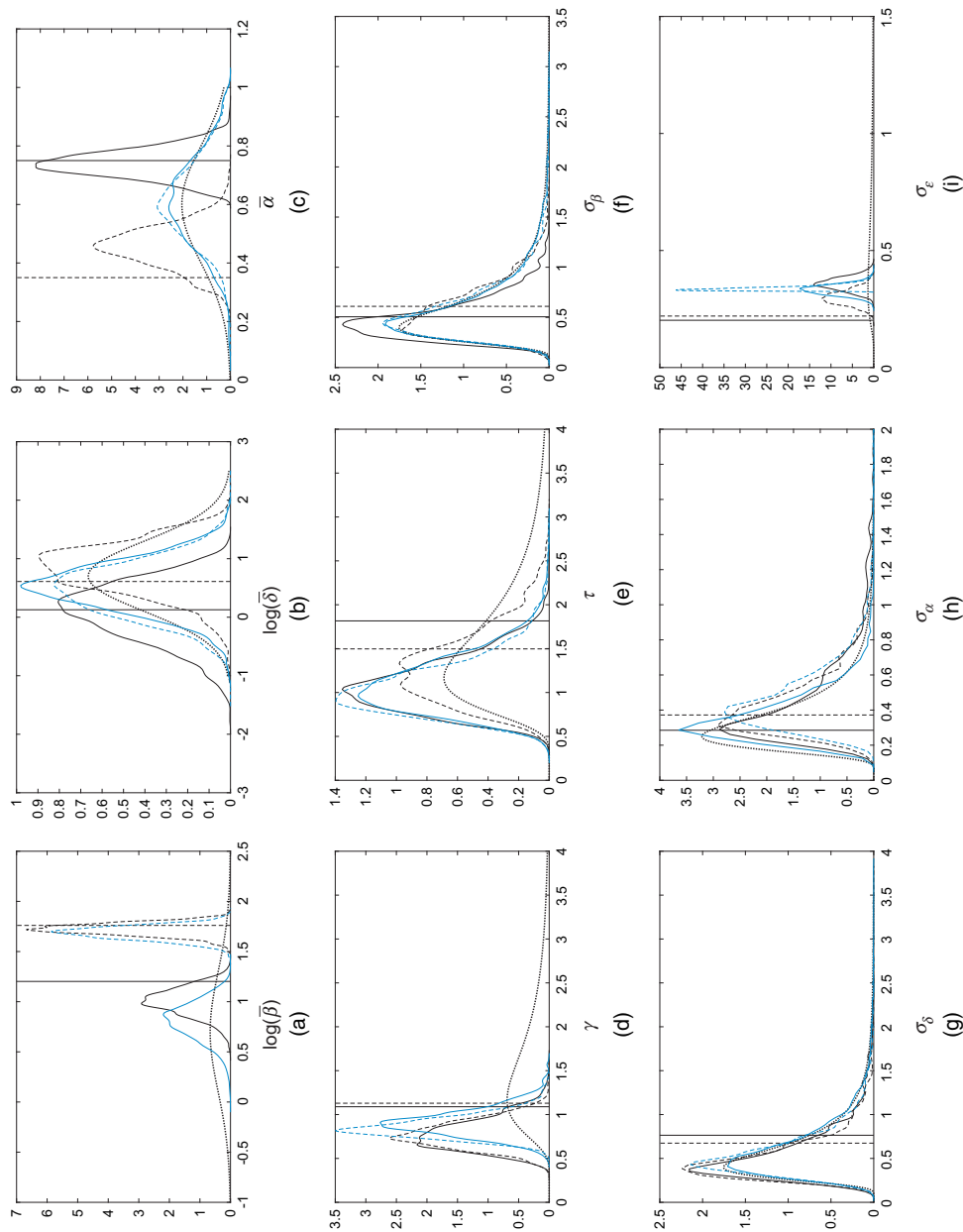


Fig. 10. Posteriors based on simulated data with $M = 17$ subjects $\mathcal{D}_1$ (---, ---) and $\mathcal{D}_2$ (---, ---) obtained with BSL (---, ---) and the PMM (---, ---) (---, ---); ground truth parameters for $\mathcal{D}_1$; (---, ---), prior densities: (a) $\log(\beta)$; (b) $\log(\delta)$; (c) α ; (d) γ ; (e) τ ; (f) σ_β ; (g) σ_δ ; (h) σ_α ; (i) σ_ϵ

variability). We considered two different SDEMEMs: one for unperturbed growth, modelling an untreated control group, and one for tumour (re)growth following an active treatment such as chemotherapy or radiation therapy. The former is a one-compartment model whereas the latter is a two-compartment model. The two compartments represent the unobserved fraction of tumour cells that has been killed by the treatment, V^{kill} , and the unobserved fraction that has survived the treatment, V^{surv} . Hence the model extends the classical double-exponential model by including random perturbations in the growth dynamics.

Parameter inference for the SDEMEM is difficult for several reasons. One is the intractability of the likelihood function. Another reason is that model parameters are difficult to identify, since data consist of noisy measurements of the total tumour volume $V = V^{\text{surv}} + V^{\text{kill}}$, not the separate compartments. Finally, most tumour xenograft studies are performed with small sample sizes. We considered methods for exact and approximate Bayesian inference to overcome this. In particular, we compared approximate Bayesian inference using the SL approach, BSL, with exact Bayesian inference using a PMM. BSL bases inference on the likelihood function of normally distributed summary statistics, instead of the intractable likelihood function of the actual data. The efficiency of the resulting estimator relies on the choice of the summary statistics. For the application to SDEMEMs we advocated the use of subject-specific summaries, which can be further comprised over groups by taking averages and computing covariances. In an application to a tumour xenography study with two active treatment groups and an untreated control, comprising data from 5–8 subjects in each group, we found that inference results produced by BSL are similar to those from a PMM, indicating that our choice of summary statistics was appropriate. The small sample bias was similar between the two methods. A further advantage of synthetic likelihoods is that, unlike exact particle-based inference, it can be applied to models other than the state space type; thus inference could be extended to other stochastic growth rate models than state space SDEMEMs. Also, results obtained with BSL are quite robust to the simulation set-up (i.e. to the number of simulated data sets per Markov chain Monte Carlo iteration), whereas our specific implementation of the PMM seems much more sensitive to changes in the number of particles considered (see the on-line supplementary material). Improvements with the PMM could be achieved with further research on simulating random effects conditionally on data.

A finding from the case-study was that larger sample sizes are needed to identify all model parameters and to obtain accurate estimates of the treatment contrasts. This was confirmed in a simulation study considering eight and 17 subjects, where BSL could identify treatment contrasts with 17 subjects in each group whereas the PMM did not.

We compared the fit of the SDEMEM for the two treatment groups in the case-study with that of an ODEMEM, assuming no within-subject variation in growth and elimination rates over time. The ODEMEM appears to fit the case-study data well, but parameters have larger uncertainties and the residual variation is much larger than for the SDEMEM. Unfortunately, the small sample sizes prevent us from determining which model is truly the better. Hence, overall realistic modelling of tumour growth dynamics in response to treatment remains an open question. Although we recommend larger sample sizes for obtaining valid statistical inference, Bayesian inference may still be used to perform exploratory analyses in small-scale experiments. In the latter case, judicious informative priors based on subject matter expertise may compensate for the otherwise too small sample size.

In conclusion, SDEMEMs enable mechanistic modelling of tumour growth and response to treatment including natural sources of variability. These may be useful for power calculations and optimal design, even in studies where more robust statistical methods are preferred for confirmatory data analysis.

Acknowledgements

We are grateful for fruitful comments by three reviewers and the Associate Editor. The research was partially supported by the Swedish Research Council (Vetenskapsrådet grant 2013-05167).

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Supporting information

Additional 'supporting information' may be found in the on-line version of this article:

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<https://doi.org/10.1017/S0007122618000050>